The four topics related with web based Information Management

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1. Dublin-core has been a metadata standard in the library world for a long time. Please explain the relationship between Dublin-core and XML, and describe how XML is applied in this metadata standard, using example(s).

The Dublin-Core Metadata Initiative began in 1995 with a workshop in Dublin, Ohio, presented by the professionals from many areas. Dublin Core, actually, is one of the standards with bigger influence in metadata area. The other metadata standards have the different focus in this area, such as AACR2, TEI (Text Encoding Initiatives headers), EAD (Encoded Archival Description), VRA CC, and so on.

DC (Dublin Core) involves many abstract concepts, including record, description set, description (about the URI), statement, property URI, value URI, and value string. DC contains mainly 15 elements as simple source description, but not limited to them. Until now, there are more than 50 other properties approved by DCMI. From outer layer to inner layer, there is a fixed order or sequence to take effect, which is record, description set, description (about the URI), statement, and most inner part, including property URI, value URI, and value string (Powell, 2006).

XML definitely is the first choice of DCMI. Actually, DC can be separately embedded into HTML, XML, and RDF in reaching different targets during the process of describing the documents (Hillmann, 2005). Because XML is the perfect bridge between different information systems, such as libraries, e-business, or some government institutes, XML definitely becomes the first choice of DCMI. Although HTML can also

undertake the similar role which XML do, for being a fixed markup language, it is merely restricted in homogeneous information systems. However, under the complicated heterogeneous circumstances, undoubtedly, XML become an indispensable choice to realize DC for its flexibility and extendibility. Furthermore, because the systems employed by different libraries, have big difference from historical reason, libraries have the emergent needs or requirements to share the bibliographic and digital information among them. Hence, XML becomes the definite choice to realize the DC standard in libraries.

The example of XML applied to DC metadata standard:

XML is used to describe the documents by applying the DC metadata standard, demonstrated as below. In order to demonstrate the entire landscape, 15 simple Dublin core metadata element set (DCMES) are totally employed.

```
<?xml version="1.0"?>
<metadata
  xmlns="http://www.ukoln.ac.uk/metadata/dcdot/"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.ukoln.ac.uk/metadata/dcdot/
http://www.ukoln.ac.uk/metadata/dcdot/dcdot.xsd"
  xmlns:dc="http://purl.org/dc/elements/1.1/">

  <dc:title>
    titleindex
  </dc:title>

 <dc:creator>
 Andy Tang
 </dc:creator>

  <dc:subject>
   The energy books information
  </dc:subject>

 <dc:description>This documents is the detailed depiction of
 international energy development environment.
 </dc:description>
```

```
 <dc:publisher>
 ITC^Deltacom NETBLCK-ITCD-3
 </dc:publisher>

<dc:contributor>
The Johnson energy library
</dc:contributor>

<dc:date>11-20-2008
</dc:date>

 <dc:type>
   Text
 </dc:type>

 <dc:format>
   text/html
 </dc:format>

 <dc:format>
   20944 bytes
 </dc:format>

 <dc:identifier>
   http://www.berrypicking.net/left/titleindex/titleindex.html
 </dc:identifier>

 <dc:source>
 The International Energy association
 </dc:source>

  <dc:language>
  En
  </dc:language>

  <dc:relation>
  IEA11.00.23
  </dc:relation>

  <dc:coverage>
  USA
  </dc:coverage>

  <dc:rights>
  International Energy Publisher
  </dc:rights>

</metadata>
```

2. “Your public library periodically publishes new books purchased by the library on its website. The information can be, and has been contained in HTML web pages, and it works fine for presenting the information. Argue why XML data

format would (or would not) be a better choice for such a publicizing purpose, in comparison to HTML files."

The first, as a Meta language, by use of DTD (document type definition), XML is much more dynamic than HTML in characteristics, meaning that it allows us to create and format document markups. This character of XML make it much more flexible and easier, to publish new books into the website and change them by using application. However, HTML markup are static, restrictedly bundled, and can not be extended. So, XML has more advantages in this aspect compared with using HTML.

The second, from the standard aspects, the XML language is designed to carry the data in form of Data Island. However, the HTML language is designed mainly to display data or information. Because, sometimes, the public library requires making some statistics on books published into the internet, or integrates the data from new books, or transfers them into other libraries' system, under this situation, XML has more advantage than HTML language on handling data directly.

The third, in case of publishing books into the website, nearly everything that HTML can do, can be completed by XML. XML is not only good at carrying data, but also adept in presenting the information into the website by use of the XSLT. That is to say, compared with HTML strongpoint on presenting information on website, XML nearly have no flaw. Furthermore, XML has some data-handling advantages that HTML does not have.

So, it is obvious that, in the professional areas, especially with the potential requirements of handling data in website, XML should be considered to be chosen prior to HTML.

3. Assuming you were an e‐business (means your transactions happen on the Web) manager and you have tens of thousands of all kinds of products. In order to maintain and make use of the product information, which of the tools you would use: a relational database DBMS or XML? Justify your choice. Analyze pros and cons with your choice. Make reasonable assumptions where you feel necessary (for example, "assuming this business is domestic/international," etc.).

The e-business we handle, involves whatever international or domestic transaction, including thousand kinds of products. Under this situation, the most advantageous tools we can choose, is the RDBMS, not XML.

3.1. There are tons of pros to explain why we put RDBMS as our choice. The pros of adopting RDBMS are in the list below.

The first, RDBMS employ integrity checking to ensure the healthy connections between different tables in database. Although the tables in RDBMS is 2 dimensions in structure, the database are definitely 3 or multiple dimensions in structure for the reason of its containing countless tables with connections of each other. However, the data contained in XML is hard or difficult to build the similar integrity checking to keep data consistency.

The second, RDBMS have the minimal data redundancy compared with the XML data. In order to decrease the redundancy, RDBMS uses primary key and foreign key in

tables, on the one hand to increase search efficiency, on the other hand to decrease the redundancy.

The third, RDBMS have many healthy schemes, such as buffer mechanism, on handling the concurrent requirements coming from multiple users, such as inserting records, deleting records, joining records, and so on. XML do not have related scheme to solve the requirements happened simultaneously. And, the unique thing XML can do is let other user waiting when one user modifies the data in Data Island. From this view, XML has worse efficiency compared with RDBMS.

The fourth, RDBMS can reasonably distribute the data with different usage frequency, into the different tables, on the one hand, to increase the search efficiency, on the other hand, to decrease the search workload when facing complicated long tables. However, in most times, XML can not automatically support this function, with a very bad efficiency when confronting large dataset.

The fifth, RDBMS provides rich procedures (API) and applications to handle the data contained in tables and database. Furthermore, some of them have many flexible and recomposed characteristics to realize complicated functions. However, comparing with RDBMS, the XML data handling functions are relatively weak, sometimes very difficult to reprogram.

The sixth, RDBMS have many tools to help constructing business model in a very short time, and it can realize them in DBMS platform quickly. However, XML can not provide such kinds of tools, and even if XML provide some tools, they are hard to integrate together to take effect.

The seventh, RDBMS can implement the data communication between backbone database and the front end webpage, through adopting CGI or ASP. As for the existing technique, end users can easily handle the data contained in database through the webpage. That is to say, RDBMS can not only realize the function XML can do, but also have the advantage XML not have.

3.2. However, there are still some cons for the RDBMS compared with XML.

The first, for RDBMS, it is a very complicated process to configure the CGI to communicate between the database and website. But, on this aspect, XML is relatively easy to use, and do not require any complicated configuration.

The second, without middleware, it is troublesome to let heterogeneous RDBMS communicate among each other. However, XML, by use of DTD, is a perfect bridge to communicate between different systems.

The third, the RDBMS is too much expensive in price, although it has satisfying performance by using complicated database technique. However, XML is rather cheap on whatever hardware or software aspects, although it sacrifices some performance compared with database.

4. Compare MySQL 6's XML supporting utilities with that of Oracle as described in your textbook. What does MySQL have and not have and what suggestions you would make for MySQL for future development in terms of processing XML data?

4.1. **The utilities and functions that Oracle supports in handling XML**

4.1.1. Oracle can structurally store the data contained in XML document by use of XML schema. It means to extract data from XML document and store them into database, simultaneously discard the XML tags (Oracle, 2008).

4.1.2. Another choice, Oracle can provide, is to employ XML type datatype, and adopt an unstructured method to store entire XML documents into database field (Powell, 2006).

4.1.3. Oracle provides related function or procedure bundled with the XML datatype field, to process XML documents.

4.1.4. Oracle can create a separated or isolated XML documents from a traditional database.

4.1.5. Oracle can create or build XML documents within a database, simultaneously storing it in a XML type field (Powell, 2006).

4.1.6. Oracle can retrieve the XML contents which are contained or stored in a XML type field within a database (Powell, 2006).

4.1.7. Oracle provides the programmability on PL/SQL to handle the XML documents.

4.1.8. Oracle provides XML publisher, combining with J developer, to generate PDF file or other file format through XML documents.

4.2. The utilities and functions that MySQL 6.0 support in handling XML

4.2.1. MySQL can output database contents into XML format files which are separately stored out of the database (Stephens, 2008).

4.2.2. MySQL 5.1 can import special XML data into MySQL table columns, but definitely require using a stored procedure (Stephens, 2008).

4.2.3. From MySQL 6.03 version, MySQL employ a relatively simple implementation to import XML data into MySQL table, which is “LOAD XML…” (Stephens, 2008).

4.2.4. From MySQL 5.15, MySQL can employ XPath functions to extract some or particular contents related with one or more elements (Stephens, 2008).

4.2.5. MySQL provides some functions to directly handle the markup contained in XML fragment, such as replacing or substituting them.

4.2.6. Some of XPath operators are supported by MySQL 5.1 and 6.0, such as “/”, “*”, and “|”. However, there are still some XPath operators not supported by MySQL 6, such as “relative locators”, and “::” (Stephens, 2008).

4.2.7. MySQL permits users to employ their own variables in XPath expression.

4.2.8. Nearly equivalent with PL/SQL from Oracle, PHP give MySQL some flexibility to handle the XML documents, but with more complicacy compared with Oracle schema.

4.3. Some advice on the future of MySQL in handling XML

4.3.1. MySQL should provide some convenient GUI to handle and control the import and output of the XML documents or data, whatever in structured or unstructured form.

4.3.2. MySQL should support entire functions of XPath in future version of MySQL.

4.3.3. MySQL needs the function to directly convert structurally stored data within tables into unstructured XML ones, without involving the extra intermediary step, such as outputting them into extra XML files.

4.3.4. Although MySQL have some third party tools to help outputting data into XML format, it is definitely not enough to stop at this point. MySQL needs some integrated tools to simplify the process of whatever importing XML into database or outputting data into XML.

4.3.5. MySQL should invent a satisfying way, for the end user, to easily map the elements in XML into related database fields, or vise versa. This will simplify the process of data transfer.

4.3.6. The PHP, as a programming language, is a little inferior in functions compared with PL/SQL by Oracle, a procedural language specifically designed for handling table and database.

4.3.7. MySQL should have the ability to simultaneously search or retrieve the XML documents stored in different fields and database, a little similar with handling the data contained in data warehouse.

4.3.8. MySQL should have a function to search or retrieve the mixture of unstructured stored XML data and regular data in concurrent way, and give the search results a different mark in differentiating data origin.

4.3.9. MySQL should provide specific “commit” and “rollback” function, designed specially for XML handling.

4.3.10. MySQL should have easy-handled XML publishing tools, to make the XML easily demonstrated into the internet.

Reference

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